Problem Set 1

Due Saturday, September 12, 11:59 PM.

1. Consider the Bohr model of the hydrogen atom with the proton fixed at the origin and the electron moving in a circular orbit around the proton subject to only the Coulomb force.

   (a) Show that for an electron undergoing centripetal motion of radius r, the total energy of the electron can be written as $E = -\frac{p^2}{2m_e} = -\frac{e^2}{8\pi\epsilon_0 r}$, where ϵ_0 is permittivity of free space, p is the momentum, m_e is the mass, $-e$ is the charge of the electron. (5 points)

   (b) Assuming that an integral multiple of de Broglie wavelengths, $n\lambda$, fit into the circumference of the circular orbit, determine the quantized radii of the orbits r_n and the quantized energies E_n as a function of integer n. Write the numerical values in meters (m) for r_1 and in electronvolt (eV) for E_1. (5 points)

   (c) Write the numerical value of the de Broglie wavelength for $n = 1$ and point out in one sentence what it means that it is comparable to the size of the atom. (5 points)

   (d) Determine the ratio of the speed of the electron in the n^{th} orbit to the speed of light, and write the numerical value for $n = 1$. Do you think relativistic effects would be important? (5 points)

   (e) Determine the angular momentum of the electron in the n^{th} orbit and show that it is an integral multiple of $\hbar$. (5 points)

   (f) According to classical electromagnetism, an accelerating charged particle radiates and loses energy at the rate given by the Larmor formula $\frac{dE}{dt} = -\frac{q^2a^2}{6\pi\epsilon_0 c^3}$, where q is the charge and a is the acceleration of the particle. If the electron initially in the $n = 1$ orbit were to radiate at the rate given by the Larmor formula, determine how long it would take for it to spiral and fall into the proton? What does this tell us about Bohr's model of the hydrogen atom with the electron orbiting around the proton? (5 points) [You may take the radius of the proton to be negligible compared to the radius of the $n = 1$ orbit.]

2. The peak frequency ω_{peak} of the radiation from a black body follows from Planck's spectral density $u(\omega) = \frac{\hbar\omega^3}{\pi^2 c^3}\frac{1}{e^{\frac{\hbar\omega}{k_\beta T}}-1}$ and $\frac{du(\omega)}{d\omega} = 0$. Solving this numerically, the peak frequency is given by $\omega_{\text{peak}} = \frac{2.82 k_\beta T}{\hbar}$.

   (a) Assuming that the radiation from the sun can be roughly modelled as that from a black body emitter at surface temperature of 5780 K, determine the peak frequency of the emitted

radiation. What is the corresponding wavelength? In which range of the electromagnetic wave spectrum does it belong? (5 points)

(b) The peak wavelength for $u(\lambda) = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{\frac{hc}{k_B T \lambda}} - 1}$ is given by $\lambda_{\text{peak}} = 0.20 \frac{hc}{k_B T}$. Determine the peak wavelength for the sun at a surface temperature of 5780 K. Why is this different from the wavelength you obtained in part (2a)? (5 points)

(c) Taking the surface of the earth as a black body emitter at temperature 288 K, determine the peak frequency of the emitted radiation. In which range of the electromagnetic wave spectrum does it belong? (5 points)

(d) The cosmic microwave background radiation (left over from the early universe) has a black body spectrum at a temperature of 2.73 K. What is the peak frequency and in which range of the electromagnetic wave spectrum does it belong? (5 points)

3. Read the note on the photoelectric and the Compton effects available here and write a one paragraph summary of the main points for each. (5+5=10 points)

4. Stop by office hours on Tuesday (9/8) 1:30 - 3 PM or Thursday (9/10) 3 - 4:30 PM in Lyman 329 and introduce yourself.

5. How long did it take you to complete this problem set?